Team 20: Generation for Recommendation

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Milestone 2

1. Methodology

Our current pipeline combines embedding-based user profiling, LLM-based prompt construction, API-based music generation, CLAP-based reranking, and lightweight visualization in an interactive demo interface. In Milestone 2, we implemented the full pipeline and ran a working demo using user `007XIjOr` as an example.

1.1 Data and User Embedding

To represent musical preference, we rely on CLAP-based song embeddings. Each track is mapped into a shared embedding space, allowing songs to be compared through vector similarity. From the retrieved songs, we aggregate genre, tag, artist, and audio-feature information to form a structured listener summary. In the demo, this profile is presented through interpretable fields including style keywords, top genres, top tags, representative artists, representative tracks, and recent listening statistics such as danceability, energy, valence, and tempo.

1.2 User Profile Construction, Listener Description, and Prompt Generation

We build a listener representation from historical user–song interactions by retrieving a set of recently listened tracks and summarizing them into a profile artifact. In the current demo, profile construction is controlled through parameters such as retrieval top-k, summary top-n, and an option to exclude recently listened songs from downstream recommendation-style retrieval. This design helps the system focus on a user's broader stylistic preferences rather than simply repeating tracks the user has just heard.

After constructing the structured profile, we use an OpenAI language model, shown in the demo as `gpt-5.3-mini`, to convert the retrieved musical evidence into both a natural-language listener description and a generation-ready prompt. Rather than prompting the music model with only raw tags, the LLM produces a more coherent musical specification that includes mood, instrumentation, vocal style, arrangement style, and production texture. This step therefore yields both a machine-readable user representation and a human-readable explanation of the user's taste.

1.3 Music Generation

For music generation, we connect the pipeline to a **third-party Suno API service**. In the current demo, one run is performed for a selected user through multiple generation calls. In the example shown, five API calls produced ten candidate tracks, which were then passed to the reranking module. Each returned candidate includes generated audio, artwork, lyrics-related text, metadata, and the exact prompt used for generation. This setup decouples personalization from synthesis: the upstream profiling module determines what kind of music should be generated, while Suno handles the actual audio generation.

1.4 CLAP-Based Reranking

Because direct music generation can be noisy, we add a post-generation reranking stage. All generated candidates are embedded again using CLAP, and each candidate embedding is compared with the target user representation using cosine similarity. Candidates that are closer to the user embedding are considered better aligned with the listener's preferences.

1.5 Interactive Demo

To make the full pipeline inspectable, we implemented a Streamlit-based demo interface. The demo integrates the main stages into a single workflow in which the user can select a known user ID, load or rebuild the user profile, generate candidate songs, rerank the candidates, and inspect selected results together with their metadata.

1.6 Evaluation Setup and Preliminary Automatic Metrics

Our evaluation pipeline is designed as a preliminary automatic assessment framework built on the same CLAP embedding space used for user profiling and reranking. For each generated clip, we compute its cosine similarity to the **user embedding**, to the centroid of the **user's recently k listened songs**, and to each **individual reference track**. From these comparisons, we derive alignment metrics such as mean similarity to the reference set, maximum similarity to the closest historical song, and top-k average reference similarity.

We then aggregate these values separately for the full candidate pool and for the reranked selected outputs, allowing us to assess whether reranking improves personalization quality. In addition, we include a simple imitation check by flagging generated clips whose similarity to the nearest reference track exceeds a configurable threshold. To evaluate diversity, we compute pairwise and nearest-neighbor cosine similarities within both the candidate set and the final selected set. Finally, we visualize the relationship among recent listened tracks, generated candidates, selected songs, and the recent-listening centroid in a two-dimensional projection using UMAP when available, or PCA otherwise.

2. Results

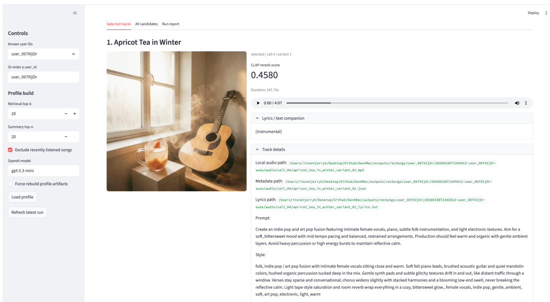

Figure 1: Demo interface example 1.

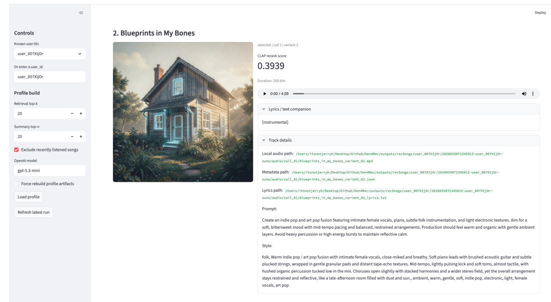

Figure 2: Demo interface example 2.

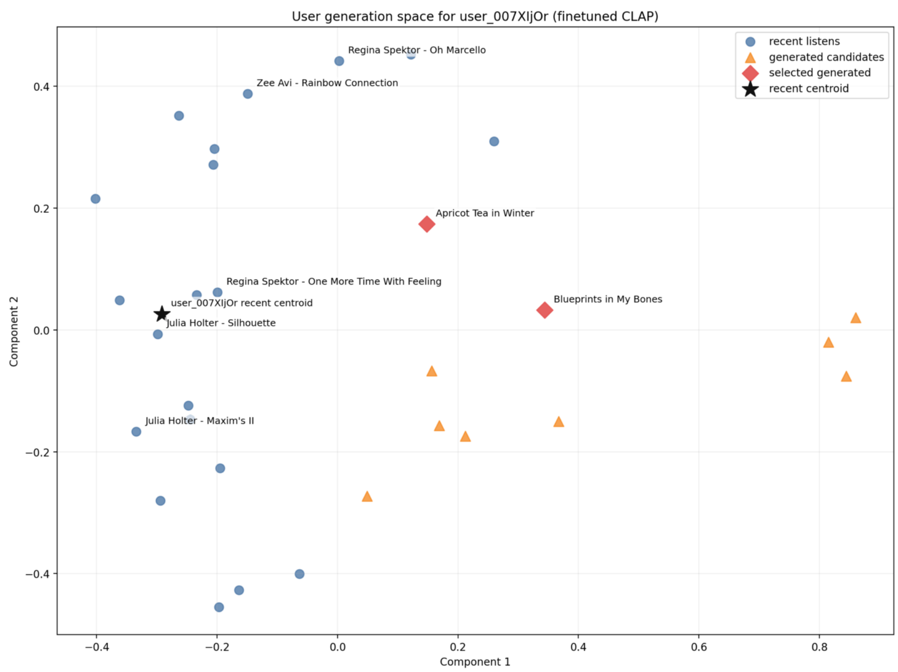

Figure 3: Embedding visualization for user 007XIjOr.

2.1 Example User Study

To further evaluate personalization quality, we conducted a user study on a real user.

We generated music based on the user's recent listening history (10 songs) and compared outputs using both human evaluation and embedding-based reranking.

From three generation attempts, one successful run produced two candidate songs. The user selected one preferred track, which was consistent with the system's reranking result.

Quantitative evaluation (on a 1–5 scale) is shown below:

- Song A: Preference Match = 5, Sound Quality = 4, Creativity = 3
- Song B: Preference Match = 5, Sound Quality = 4, Creativity = 4

The user preferred Song B due to its distinctive style, although both songs aligned well with their preferences.

These results suggest that:

1. The system is able to generate music that matches user preferences effectively.
2. CLAP-based reranking aligns with human judgment.

However, several limitations were observed:

- Generated profiles tend to summarize known preferences rather than uncover deeper or novel insights.
- Some generated content lacked vocal elements, likely due to training data bias toward instrumental music.
- Profile generation occasionally includes artists not directly listened to by the user, due to reliance on retrieved similar tracks.

3. Analysis

The project now demonstrates a coherent end-to-end pipeline: listening history and embeddings are used to generate LLM-based listener profiles, support music generation, and enable CLAP-based reranking. This pipeline is further supported by a prototype interface that allows the team to inspect prompts, audio outputs, and intermediate artifacts. A major strength of the current system is its transparency. Because the workflow is organized through scripts and saved artifacts, team members can debug, refine, and compare outputs without depending on a black-box process.

At the same time, several limitations remain. The system is still sensitive to prompt wording, negative constraints, and template design, meaning that small prompt changes can materially affect generation quality and personalization outcomes. In addition, claims of improved personalization remain difficult to validate unless evaluation is carefully defined and aligned with human judgment. Current constraints also include reliance on a fixed dataset, dependence on external APIs, and the broader limitation that embedding-space similarity does not fully capture subjective musical preference.

Accordingly, the highest-priority work for this stage falls into two areas. First, the team must develop a stronger evaluation framework with clear metrics, baselines, and comparisons, including zero-shot versus fine-tuned CLAP. Second, the team should continue improving prompt engineering for both generation quality and personalization by testing template variants, instructions, and constraints. These two efforts should reinforce one another: evaluation results should guide prompt revisions, while more stable prompts should make experiments more comparable before the team shifts greater attention to broader UI polish.

4. Plan for Additional Analysis

Our next phase will focus primarily on strengthening evaluation. While our current automatic metrics based on CLAP similarity provide a useful first-pass measure of alignment, they do not fully capture perceived musical quality, diversity, or listener satisfaction. We therefore plan to refine the evaluation framework by separating multiple dimensions of performance, including personalization alignment, diversity among generated outputs, and possible over-imitation of historical tracks.

Because music preference is inherently subjective, we also plan to complement automatic evaluation with human evaluation. In particular, we aim to collect listening-based judgments on whether generated outputs match the user's style, sound musically coherent, and remain sufficiently diverse. This will help us better understand where embedding-based metrics are informative and where they fail to reflect human perception.

In addition, we plan to move beyond a single demo case and run experiments on a larger subset of users. This will allow us to test whether our reranking strategy and prompt design generalize across different listener profiles rather than only working for one selected example.

Finally, we plan to improve and compare prompt engineering strategies. Our current pipeline uses a relatively baseline prompting setup in which the LLM generates prompts largely from the structured listener profile. However, music generation models such as Suno may respond better to more structured and model-aware prompt formats. We therefore plan to compare alternative prompt templates, instructions, and constraints to determine which formulations produce the best balance of quality and personalization. If time permits, we may also test whether the same personalization pipeline can transfer to a different music generation backend, which would help assess the robustness of our approach.

5. Work Plan

5.1 Contributions So Far

Tony: Built the user and song embedding pipeline, integrated tags, genres, and metadata into the downstream profiling workflow, and organized the repository and data pipeline so the team can run the system consistently.

Conny: Leads the LLM profiling stage, transforming retrieved music features into structured user profiles and prompts, while also driving evaluation design and early validation metrics.

Jerry: Runs the model to generate embedding `.npy` files, leads the music generation stage, and builds the reranking pipeline and Streamlit demo interface for showcasing prompts, generated audio, and results.

5.2 Tentative Weekly Plan

Week 10

- Refine prompt engineering for generation quality and personalization.
- Test prompt template variants, negative constraints, and instruction wording.
- Finalize the evaluation setup and scoring rubric.

Week 11

- Run evaluation experiments on a small subset of users.
- Compare original CLAP versus fine-tuned CLAP in reranking.
- Compare prompt variants based on alignment, quality, and diversity.

Week 12

- Analyze evaluation results and identify which factors improve personalization most.
- Improve environment setup and reproducibility across team members.
- Debug dependency, configuration, and API issues.

Week 13

- Finalize the Streamlit demo and polish the pipeline.
- Refine the main `README`.
- Complete supporting documentation, including `cli.md` and `streamlit.md`.

Milestone 3 Deliverable (5/3)

- Prepare the final presentation flow and supporting materials.

Appendix A.

In this section, we present the objective evaluation metrics used from Section 1.6:

Generation Embedding: Let $g \in \mathbb{R}^d$ denote the embedding of a generated audio clip in the fine-tuned CLAP space, with $\|g\|_2 = 1$.

User Embedding: Let $u \in \mathbb{R}^d$ denote the user embedding, constructed from the user's listening history, and let $\mathcal{R}_u = \{r_1, r_2, \ldots, r_M\}$ be the set of M recent reference-track embeddings for user u, where each $r_i \in \mathbb{R}^d$ is also L2-normalized.

User-embedding alignment. We measure how well a generated clip aligns with the overall user representation using cosine similarity:

$$S_{\text{user}}(g, u) = g^\top u. \tag{1}$$

Recent-centroid alignment. We define the centroid of the user's recent reference tracks as

$$c_u = \frac{1}{M} \sum_{i=1}^{M} r_i, \tag{2}$$

and normalize it as

$$\hat{c}_u = \frac{c_u}{\|c_u\|_2}. \tag{3}$$

The centroid-based personalization score is then

$$S_{\text{centroid}}(g, u) = g^\top \hat{c}_u. \tag{4}$$

Reference top-k alignment. To better capture multi-modal user taste, we also compare a generated clip against the user's recent reference tracks individually. For each reference track r_i, we compute

$$s_i = g^\top r_i. \tag{5}$$

Let $s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(M)}$ denote these similarities sorted in descending order. We then define the reference top-k personalization score as

$$S_{\text{ref-top}k}(g, u) = \frac{1}{k} \sum_{j=1}^{k} s_{(j)}. \tag{6}$$